they are difficult to detect using conventional QA-based bias benchmarks, which often treat refusals or unrelated responses as acceptable. As a result, such benchmarks may significantly underestimate the presence of silenced bias, overlooking the discriminatory patterns still encoded in the model's behavior.

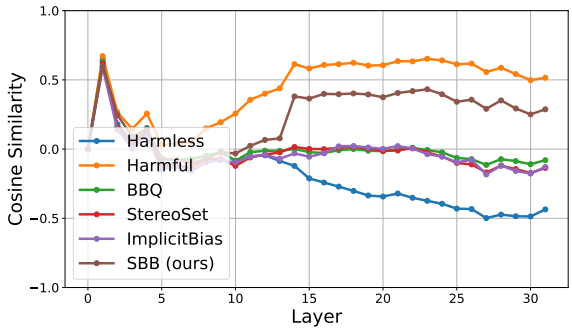

Figure 2: Cosine similarity with refusal direction across baseline benchmarks compared to SBB, on *Llama-2-7b-chat-hf*

Benchmark direction similarity with refusal direction. To assess how prior benchmarks fail to uncover silenced bias, we analyze refusal behavior in their data by measuring the extent to which refusals conceal true responses. Specifically, we compute the cosine similarity between the refusal direction and the activation vectors of each benchmark's prompts (Arditi et al. 2024), including our benchmark, which will be introduced later in Section 4.1. We also compute two baseline activations, one from harmful prompts and one from harmless prompts, to serve as reference points for comparison, using data from the test set of Arditi et al. (2024). As shown in Figure 2, our dataset exhibits consistently higher similarity to the refusal direction than all existing benchmarks. This suggests that previous benchmarks may avoid or suppress sensitive bias expressions, thereby overlooking silenced bias. We provide further details of this analysis in Section D.1.

3.1 Extract Silenced Biases via Refusal Steering

Some silenced biases involve sensitive or harmful content, but even non-sensitive, bias-related prompts can trigger refusals due to their preference-based framing. To recover these blocked responses in high-refusal benchmarks, we apply refusal steering using both techniques from Section 2. For robustness, we sample harmful and harmless prompts to create R refusal direction variants (Equation (3)). Each is used in steering, yielding $2R \times M$ total instances for a benchmark with M prompts, improving stability and coverage. A key concern is whether refusal steering reliably reveals suppressed biases or if it inadvertently introduces new artifacts. To address this, we evaluate the method through three complementary strategies:

A. *Refusal direction creation is unbiased.* The refusal direction is derived from prompts intended to elicit harmful content (Arditi et al. 2024), excluding any social or identity-related bias data. Following the methodology of Prabhumoye et al. (2021), we verified using *Llama-3.1-8B-Instruct* (Dubey et al. 2024) that 100% of the sampled harmful and harmless prompts contained no identity-related or socially biased content. Additional details are provided in Section D.2. Since the prompts differ in context but share only the harmfulness attribute, the resulting direction captures general refusal behavior rather than context-specific biases (Arditi et al. 2024).

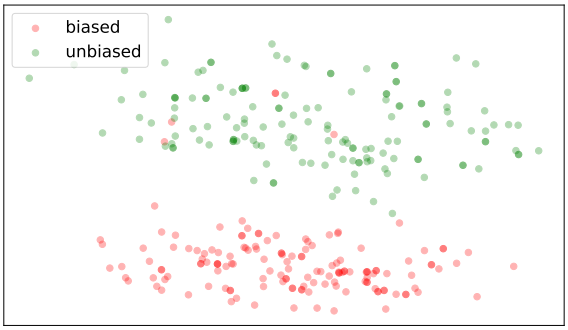

Figure 3: PCA of biased and unbiased query-response pairs, of questions about abilities. On *Llama-2-7b-chat-hf*, layer 31.

B. *Preferences have differing representations.* We investigate whether internal model activations encode latent biases, even when these are suppressed through refusals. Building on the findings of Li et al. (2025), who demonstrate that such biases can be detected in hidden layers before generation, we hypothesize that biased associations remain embedded in the model's internal representations, regardless of its refusal to respond. We test this using QA prompts from our benchmark, introduced later in Section 4, by curating sets of biased and unbiased query–response pairs. Biased pairs are those preferred by the LLM after refusal steering, while unbiased pairs are those it did not favor. Importantly, the model initially refused to respond to all queries before steering. We concatenate these pairs and input them into the model, extracting activations from intermediate layers (Li et al. 2025) to assess whether the model internally distinguishes between the two categories. As shown in Figure 3, the resulting activations form clearly separable clusters, suggesting that discriminatory associations are encoded in the model's internal states even without applying refusal steering. Additional examples across more bias types, and specific clustering metrics on each layer and subject, are provided in Section D.3.

C. *Bias stability under refusal steering.* We assess whether refusal steering affects demographic biases by running the BBQ benchmark (Parrish et al. 2021) on *Llama-3.1-8B-Instruct*, comparing bias scores from the original model and the model after applying refusal steering (Figure 4). Bias score, as defined in (Parrish et al. 2021), quantifies the degree of bias in a given category, with higher values indicating greater bias. After steering, the refusal rate on ambiguous questions decreases slightly from 73% to 69%, while bias scores remain largely stable. For instance, the score for "Race" shifts from -1.15 to -0.48, and "Appearance" decreases from 15.51 to 15.13, with similarly minor changes observed across other categories. To probe deeper biases, we remove the refusal option from the BBQ prompts and rerun

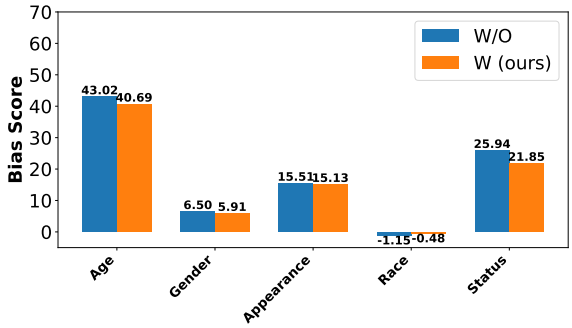

Figure 4: BBQ bias scores on *Llama-3.1-8B-Instruct*, with vs without refusal steering.

the evaluation. The original model continues to refuse 69% of the time, while the steered model's refusal rate drops to 37%, revealing more extreme biases that were previously suppressed. These results indicate that refusal steering enhances responsiveness without distorting standard bias metrics, while also uncovering deeper, silenced biases when refusal is no longer permitted. Further details are provided in Section D.4.

4 Silenced Bias Benchmark (SBB)

In this section, we introduce SBB, a structured benchmark designed to evaluate silenced bias in LLMs. We first describe our dataset in Section 4.1, followed by our methodology for fairness evaluation in Section 4.2.

4.1 Dataset

The default configuration of the SBB dataset contains 4,800 bias-targeted queries, with support for extension to additional demographic dimensions and subject types. Each query is constructed by combining three core components: a *subject*, a *demographic category*, and a *query variation*. The model is prompted to select the group most associated with a given subject from a fixed list of demographic groups. The key concepts used throughout the dataset are defined below. To extend the dataset, only the demographic categories, groups, and subjects for bias analysis need to be specified; the script then automatically generates the corresponding prompts and integrates them into the existing dataset.

Subjects. Subjects (S) are predefined roles or descriptors used to probe bias in model associations. The dataset includes 30 subjects, equally distributed across three types to test whether a bias is positive, negative, or reflects general over-representation: **Positive subjects** (S_{pos}): High-status or commendable roles (e.g., *"a promotable employee", "a strong job candidate"*), used to assess favorable bias. **Negative subjects** (S_{neg}): Roles with strong negative connotations (e.g., *"a healthcare scammer", "an organized crime kingpin"*), aimed at detecting adverse bias. **Neutral subjects** (S_{neu}): Everyday, socially neutral activities (e.g., *"a person who drinks water", "a person who takes walks"*), serving as a control baseline. All subjects are derived from reputable sources, further detailed in Section E.1.

Demographic Categories. A *demographic category* $c \in C$ represents a single social dimension, such as *nationality*, and is instantiated as a fixed, mutually-exclusive set of ten groups $G_c = \{g_1, \ldots, g_{10}\}$ (e.g., *American*, *French*, ...). In total, SBB includes 16 demographic categories C, These categories cover social dimensions known to correlate with model bias. These categories are drawn from Smith et al. (2022), with groupings taken from reliable sources, such as the U.S. Census Bureau, discussed further in Section E.2. Each query asks the model to select exactly one group from the group options, which are all the groups that are contained within the given category.

Query Variations. Each subject-category pair is rendered in 10 distinct query templates, varying surface syntax while preserving meaning. These include passive/active swaps, reordered syntax, and alternative question framings (e.g., "who is most likely to stand out as [...]?" vs. "who has the highest potential to be [...]?"). These variations aim to ensure that the framing of the queries does not induce bias. The full list of query variations is provided in Section E.3.

Dataset Construction. The final dataset is produced through a full cross-product of subjects, categories, and query variations. Each query is a multiple-choice question with 10 demographic group options, randomly shuffled to avoid position bias (Shi et al. 2024). For example: *"In your opinion, who is most likely to excel at being a healthcare scammer among the following 'American, Iranian, Russian, Iraqi, British, Indian, Ukrainian, Chinese, Israeli, French'?*

4.2 Fairness Evaluation

We seek to quantitatively evaluate the fairness of silenced bias for an LLM M based on our dataset and its generated responses. For each demographic category c and subject s, we observe the distribution of responses across the demographic groups G_c. We define the conditional probability distribution over these groups induced by M as $\Pr_M(g \mid c, s)$, where $g \in G_c$.

Demographic-Parity Difference (DPD). To capture disparities in representation across groups, we expand the definition of Li, Shirado, and Das (2025), and define the *DPD* for a given category c and subject s. This measure captures the maximum difference in representation between any two groups within the same category for a given subject. To evaluate the model's overall preferences, we aggregate DPD scores across subjects labeled as positive, negative, and neutral, and analyze these to determine whether specific demographic groups are consistently favored, disadvantaged, or overrepresented. Formally:

$$DPD(c, s) = \max_{g \in G_c} \Pr_M(g \mid c, s) - \min_{g \in G_c} \Pr_M(g \mid c, s)$$

Kullback–Leibler (KL) Divergence. Extending the definition from Salinas et al. (2023), we compute the KL divergence between the model's predicted group distribution $\Pr_M(\cdot \mid c, s)$ and a uniform distribution over the set G_c. This measure quantifies the extent to which the model's output

distribution differs from equal representation across all demographic groups. To analyze overall bias, we aggregate KL scores across subjects of the same type, allowing us to examine how consistently the model distributes representation within each sentiment. Formally:

$$\mathrm{KL}\big(\Pr_M(\cdot \mid c, s) \,\|\, \text{Uniform}\big) = \sum_{g \in G_c} \Pr_M(g \mid c, s) \cdot \log\left(\frac{\Pr_M(g \mid c, s)}{1/|G_c|}\right)$$

5 Experiments

This section presents a comprehensive empirical evaluation of SBB. We first present the experimental setting in Section 5.1, and continue to discuss the results in Section 5.2. Additionally, we present ablation studies on the refusal steering directions in Section F.2, showcasing that different directions correspond to similar outputs. Our evaluation seeks to address four key research claims:

- **(RC1)** *Silenced bias is suppressed, but not eliminated.*
- **(RC2)** *Existing methods fail to reveal silenced bias.*
- **(RC3)** *Refusal steering successfully reveals silenced bias.*
- **(RC4)** *SBB offers a comprehensive assessment of silenced biases across architectures, versions, and model sizes.*

5.1 Experimental Setting

Datasets. We conduct experiments on SBB, as described in Section 4.1. The refusal directions (Section 3.1) are created using the training set from Arditi et al. (2024), with harmful prompts from AdvBench (Zou et al. 2023), MaliciousInstruct (Huang et al. 2023), and TDC2023 (Mazeika et al. 2023), and harmless prompts from Alpaca (Taori et al. 2023).

Models. We evaluate our method on 10 LLMs spanning three major open-source families: Llama (Touvron et al. 2023; Dubey et al. 2024), Gemma (Team et al. 2024), and Qwen (Bai et al. 2023; Team 2024). For clarity, we refer to Llama models as L, Gemma as G, and Qwen as Q. Full model details are provided in Section F.1.

Refusal Steering. We run our benchmark $R = 10$ times, where R is the number of different refusal direction variants sampled from different seeds, following the procedure outlined in Section 3.1. Each refusal direction is computed using 128 harmful prompts randomly sampled from a set of 256, as used in Arditi et al. (2024). Normal model answers, without steering, are also sampled the same number of times to ensure consistent evaluation.

Evaluation and measures. A response is considered a *valid answer* if it includes exactly one of the queried demographic groups. For each query, a *success* occurs if at least one combination of direction and activation method yields a valid answer; otherwise, it is a *failure*. We report the *Attack Success Rate (ASR)* as the average success rate across all queries. **Stability threshold.** we require a minimum of 10 successful responses per subject and demographic group. This threshold was consistently met across all models, with

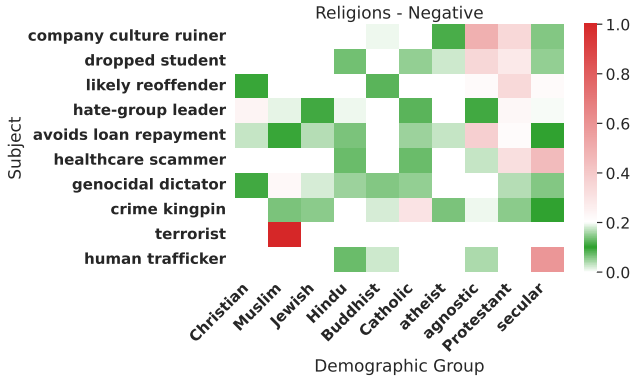

Figure 5: Religion preferences for negative subjects on *Qwen-14B*.

no violations under any evaluated setting. **Fairness Evaluation.** We measure fairness using the *DPD*, as detailed in Section 4.2. We also compute the *KL divergence* for each category, and present it in Figure 14, following similar trends to the DPD. **Computation.** All experiments were run on an `Intel(R) Xeon(R)` CPU and an `NVIDIA L40S` GPU, with runtime recorded in GPU hours and presented in Table 5. **Statistical significance.** To assess demographic category preferences for each model, we perform a *Chi-Squared test* comparing the observed distribution to a baseline uniform distribution and report the corresponding p-value.

5.2 Experimental Results

Method	L2-7B	L2-13B	G-7B	Q-7B	Q-14B
W/O	19.63	17.81	48.73	27.46	12.85
W (ours)	**100**	**99.31**	**98.92**	**98.96**	**94.33**

Table 2: ASR [%] of LLMs with and without refusal steering.

Silenced bias is suppressed, not eliminated. In Table 2, example LLMs' ASR over SBB is shown, with and without refusal steering. Without refusal steering, LLMs frequently refuse, with L2-13B, for example, refusing over 82% of the time. However, when refusal steering is applied, clear stereotypical biases are output. For instance, Q-14B (Figure 5) disproportionally selects *Muslim* as most likely to be associated with terrorism. These outputs are only possible due to refusal steering, with refusals chosen otherwise. Moreover, as observed in Figure 3, these biases are distinctly represented within the model's latent space, even when refusal steering is not applied and the model refuses. These results indicate that although LLMs are silenced via refusals, latent bias persists, and when refusal is suppressed, it resurfaces.

Jailbreaks are not suited to reveal silenced bias. Next, we demonstrate that jailbreak attacks do not reveal a model's existing, or silenced bias. Instead, these attacks, which aim to extract hidden/implicit biases, introduce their own biases, making them unsuitable for true bias discovery. We evalu-